

Module 2: Network + Network Security , Maintenance and Troubleshooting

Network Threats and Mitigation and Physical Security Risk

Task 1:

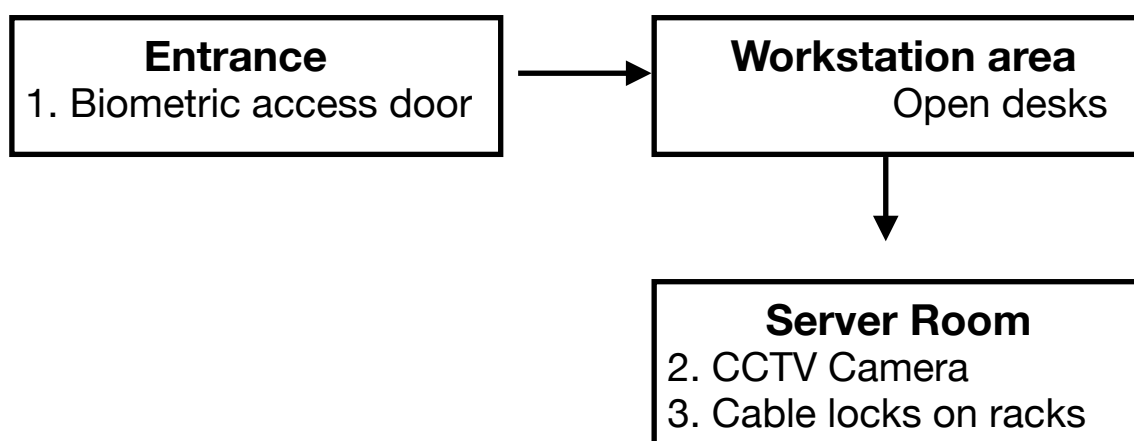
Threat	What it does	Use Case
Phishing	Tricks users into revealing credentials via fake messages/links	Fake "Your Paytm KYC is expiring, click to verify" SMS/WhatsApp messages that link to a lookalike Paytm login page
DDoS (Distributed Denial of Service)	Floods a server with traffic to make it unavailable	Ticketing platforms like BookMyShow or IRCTC occasionally facing traffic spikes/attacks during high-demand sales that slow or crash the site
Man-in-the-Middle (MITM)	Intercepts communication between two parties on an untrusted network	Someone on unsecured public WiFi intercepting an unencrypted app's login data at a café (this is exactly why apps use HTTPS/TLS, and why Task 3 last time asked about VPNs)
Malware	Malicious software that damages, spies on, or takes control of a device	Fake "WhatsApp Plus" or modded APKs that users sideload, which secretly steal contacts or messages

Threat	What it does	Use Case
Password attacks	Guessing or cracking passwords/PINs through automated attempts	Attackers running credential-stuffing attacks against Instagram accounts using leaked password lists from other breached sites

Task 3:

Risk	Mitigation
Unauthorized physical access (someone walks in and tampers with servers)	Biometric or keycard access control on the server room door, with access logs
Fire damage	Install a fire suppression system (e.g., FM-200 gas-based, not water sprinklers) and smoke detectors
Power outage/surge	UPS backup plus a diesel generator, with surge protectors on all racks
Theft of hardware	CCTV monitoring plus cable locks/rack locks on individual servers

Task 4:



Task 5:

Network admin checklist for a college computer lab

1. **Enforce unique logins, not shared lab accounts** — each student/staff member gets their own credentials so activity can be traced and one compromised password doesn't expose everyone.
2. **Auto-lock idle machines after 2–3 minutes** — labs have high foot traffic and students often walk away mid-session, leaving sessions open for the next person.
3. **Restrict USB/removable media on lab PCs** — a common malware entry point in shared-use labs where anyone can plug in a random drive.
4. **Keep OS and antivirus definitions on auto-update** — lab machines are reimaged less often than personal devices, so they lag behind on patches if not automated.
5. **Physically secure the lab's router/switch in a locked cabinet, not an open corner** — student labs often have exposed networking equipment that anyone can unplug or tamper with between classes.
6. **Log and review WiFi/network access, especially after hours** — colleges often have lab WiFi accessible campus-wide, so anomalous overnight or off-hours traffic is worth flagging.
7. **Post visible reporting instructions (who to call for a security issue)** — in a college setting, the person noticing a problem is often a student, not IT staff, so they need a clear, simple way to report it immediately.